Single-Position Intervention Fails: Distributed Output Templates Drive In-Context Learning

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Abstract

Understanding how large language models encode task identity from few-shot demonstrations is a central open problem in mechanistic interpretability. Prior work uses linear probing to localize task representations, reporting high classification accuracy at specific layers. We reveal a striking dissociation: probing accuracy completely fails to predict causal importance. Single-position activation intervention achieves **0% task transfer across all 28 layers** of Llama-3.2-3B—despite 100% probing accuracy at those same positions. This null result is itself a key finding, demonstrating that task encoding is fundamentally distributed. Multi-position intervention—replacing activations at all demonstration output tokens simultaneously—achieves **up to 96% transfer** (N=50, 95% CI: [87%, 99%]) at layer 8, pinpointing for the first time the causal locus of ICL task identity. We establish the generality of these findings across four models spanning three architecture families (LLaMA, Qwen, Gemma), discovering a universal intervention window at ∼30% network depth. Causal tracing uncovers an asymmetric architecture: the query position is strictly necessary (53–100% disruption) while no individual demonstration position is necessary (0% disruption)—resolving a key ambiguity in prior accounts. Crucially, transfer depends on internal representation compatibility, not surface similarity (r=−0.05 vs r=0.31), ruling out trivial explanations. These results establish the *distributed template hypothesis*: ICL task identity is encoded as output format templates distributed across demonstration tokens, fundamentally reshaping our understanding of how in-context learning operates.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; *Neural networks.*

Keywords

In-context learning, Mechanistic interpretability, Activation intervention, Large language models, Task representations

1 Introduction

In-context learning (ICL) enables large language models to perform new tasks from demonstrations without parameter updates [1, 2], an emergent capability [3] central to mechanistic interpretability and activation steering [4].

Prior work has approached this question primarily through linear probing: training classifiers to predict task identity from activations at specific layers and positions [5, 6]. These studies consistently find high probe accuracy, suggesting that task information is readily decodable from the residual stream. However, a representation being *decodable* does not imply it is *causally relevant* to model behavior [7, 8].

We investigate the causal role of task representations through activation intervention experiments on Llama-3.2-3B-Instruct. Our central finding is surprising: **single-position intervention fails completely**. Despite probing achieving 100% accuracy at demonstration positions and 83% at the query position, replacing the activation at any single (layer, position) pair yields 0% task transfer across all 28 layers. Control experiments confirm these positions are not causally necessary—the model ignores single-position perturbations entirely, suggesting distributed, fault-tolerant encoding.

The breakthrough comes from **multi-position intervention**. When we simultaneously replace activations at all demonstration token positions, we achieve 96% task transfer for format-compatible pairs (N=50, 95% CI: [87%, 99%]; mean 16.7% across all 56 pairs). Both `all_demo` (96%) and `output_only` (94%) conditions succeed, while `input_only` yields 0%—output tokens carry the primary signal, though replacing all tokens provides marginal improvement. This works only at ∼30% network depth (layer 8 for 28-layer models) and only when source and target tasks share compatible output formats. These findings replicate across four models (Llama-3B, Llama-1B, Qwen-1.5B, Gemma-2B), with the optimal layer consistently at ∼30% depth. The layer specificity suggests a staged pipeline: early layers encode templates, middle layers aggregate information, and late layers commit to output generation.

Our main contributions are:

- **Negative result:** Single-position intervention achieves 0% transfer at all 28 layers despite 100% probing accuracy.
- **Multi-position breakthrough:** Simultaneous intervention on demo output positions at ∼30% depth achieves 96% transfer (CI: [87%, 99%]), replicating across four models.
- **Causal dissociation:** Query is necessary but not sufficient; demos are collectively sufficient but individually not necessary. Transfer scales sharply with output position count (1–3→0%, 10→10%, all→90% mean) and source demos (1–3→0%, 5→93%).
- **Predictable transfer:** Only 7/56 pairs (13%) achieve ≥50% transfer, but these form a predictable cluster of procedural tasks with similar output structures.

2 Background

2.1 Problem Setting

We study in-context learning in the standard few-shot setting. Given k demonstration pairs $\{(x_i, y_i)\}_{i=1}^{k}$ and a query input x_q, the model generates output by concatenating demonstrations and query. The model must infer the task from demonstrations alone—no task description is provided.

Our goal is to identify which (layer, position) pairs encode task identity in a causally relevant way. This requires distinguishing *decodability* (can a probe extract task information?) from *causal relevance* (does intervention change behavior?). We address the latter through activation intervention experiments.

Table 1: Task suite: procedural, numeric, and semantic.

Regime	Tasks	Example
Procedural	uppercase, first_letter, repeat_word, length	"cat"→"CAT"
Numeric	linear_2x	"5"→"10"
Semantic	sentiment, antonym, pattern_compl.	"happy"→"pos"

2.2 Prior Work

(author?) [9] identified induction heads as key ICL circuits; recent work has traced behaviors to interpretable circuits [10, 11]. Activation patching [12, 13] finds localized sites for factual recall—facts can be edited at single MLP layers. Our single-position failure suggests ICL task identity is mechanistically distinct from factual knowledge. **(author?)** [14] note the gap between decodability and causal relevance that we demonstrate: a representation being *present* does not imply it is *used*.

3 Method

3.1 Notation and Preliminaries

Following Elhage et al. [15], we denote $h_\ell^{(p)} \in \mathbb{R}^d$ as the residual stream activation at layer ℓ and position p.

ICL Prompt Structure. An ICL prompt consists of k demonstration pairs followed by a query:

$$\underbrace{[x_1 \to y_1] \cdots [x_k \to y_k]}_{\text{demos}}[x_q \to ?] \tag{1}$$

We identify three semantically distinct position types: **demo input positions** (tokens in x_i), **demo output positions** (tokens in y_i), and **query positions** (tokens after the final "Input:"). This distinction proves critical—our experiments reveal that demo output positions carry the primary task signal (`output_only` achieves 94% vs. `input_only` at 0%), though replacing all demo tokens provides marginally higher transfer (96%).

Intervention Outcomes. After intervention, outputs are classified as **transfer** (correct for source task), **preserve** (correct for target), or **malformed** (neither). Transfer rate $\tau = n_{\text{transfer}}/n_{\text{total}}$ quantifies success.

3.2 Task Suite

We evaluate on 8 tasks spanning three computational regimes, designed to test whether our findings generalize across task types:

All tasks achieve 96–100% accuracy with 5-shot prompting, confirming model competence.

3.3 Probing for Localization

We train nearest-centroid classifiers at each (layer, position) pair to identify where task information is decodable. For each task t, we compute the centroid μ_t of activations and classify by nearest centroid: $\hat{t} = \arg\min_t \|h_\ell^{(p)} - \mu_t\|_2$. We probe three positions: `last_demo_token`, `separator`, and `first_query_token`, spanning the flow from demonstrations to query.

3.4 Single-Position Intervention

Single-position intervention tests whether task identity is localized to individual (layer, position) pairs. For source task $\mathcal{T}_s$ and target task $\mathcal{T}_t$, we compute mean source activations $\bar{h}_s = \frac{1}{n_s}\sum_i h_\ell^{(p)}(\text{prompt}_i^s)$ and replace target activations: $h_\ell^{(p)} \leftarrow \bar{h}_s$. We test all 28 layers at `last_demo_token`.

Control Experiments. We run zero ablation ($h_\ell^{(p)} \leftarrow \mathbf{0}$) and random ablation ($h_\ell^{(p)} \leftarrow \mathbf{z} \sim \mathcal{N}(0, \sigma^2 I)$) to test whether positions are causally necessary. If ablation has no effect, the position is redundant.

3.5 Multi-Position Intervention

Single-position intervention replaces one activation vector; multi-position intervention replaces activations at *multiple* positions simultaneously. This tests whether task identity is distributed across positions such that no single position is sufficient but the collection is.

We define four scopes:

- `all_demo`: All tokens within demonstration pairs (both inputs and outputs)
- `input_only`: Only tokens within "Input: X" portions of each demonstration
- `output_only`: Only tokens within "Output: Y" portions of each demonstration
- `last_demo`: Only tokens within the final demonstration pair

For each position set $\mathcal{P}$, we extract position-wise mean activations from source prompts and inject them simultaneously during the target forward pass. The replacement is applied only on the initial forward pass; subsequent autoregressive generation uses the model's own activations. Our N=50 experiments show `output_only` achieves 94% while `input_only` achieves 0%, confirming task identity is encoded in outputs. We test layers 8, 12, 14, and 16.

3.6 Causal Tracing via Noise Injection

Transplantation tests *sufficiency*; noise injection tests *necessity*. We inject Gaussian noise scaled to activation variance:

$$h_\ell^{(p)} \leftarrow h_\ell^{(p)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_{\text{noise}}^2 I) \tag{2}$$

where $\sigma_{\text{noise}} = 3\sigma_{\text{act}}$ and σ_{act} is the empirical standard deviation of activations at that layer. The **disruption rate** $\delta = (\text{Acc}_{\text{clean}} - \text{Acc}_{\text{noised}})/\text{Acc}_{\text{clean}}$ measures necessity: 100% indicates complete failure; 0% indicates the position is not necessary.

3.7 Format Compatibility Analysis

We test whether transfer depends on output format by creating task variants with identical operations but different formats. Baseline controls (random source, magnitude-matched noise, shuffled positions) verify transfer requires correct task-specific content.

Table 2: Probe accuracy (%) by position and layer.

Position	L4	L12	L20
last_demo_token	100	100	100
separator	100	100	100
first_query_token	62.5	**83.3**	75.0

Table 3: Single-position intervention fails at all layers; ablation controls confirm positions are not causally necessary.

Method	Position	Layers	Transfer
Transplant	last_demo_token	0–27	0%
Transplant	first_query_token	0–27	0%
Zero ablation	last_demo_token	14	0% (100% acc)
Random ablation	last_demo_token	14	0% (100% acc)

4 Experiments

4.1 Experimental Setup

Models. Primary experiments use **Llama-3.2-3B-Instruct**[1] (28 layers). For replication: Llama-3.2-1B (16 layers), Qwen2.5-1.5B [16] (28 layers), Gemma-2-2B [17] (26 layers). We use 5-shot prompts and greedy decoding.

Activation Extraction. We extract activations from $n_s = 10$ source-task prompts and compute position-wise means: $\bar{h}_\ell^{(p)} = \frac{1}{n_s}\sum_{i=1}^{n_s} h_\ell^{(p)}(\text{prompt}_i^{\text{source}})$.

Evaluation Metrics. We report **transfer rate** τ (fraction producing source-task outputs after intervention) and **disruption rate** δ (accuracy drop under noise: $\delta = (\text{Acc}_{\text{clean}} - \text{Acc}_{\text{noised}})/\text{Acc}_{\text{clean}}$). Each experiment uses $n_t = 10$ test inputs per task pair, averaged across 56 task pair combinations.

4.2 Probing Results

Probes achieve near-perfect accuracy at demo positions across all layers (Table 2). Query accuracy peaks at 83% at layer 12, suggesting information flows from demos to query—but as we show next, high probe accuracy does not guarantee intervention success.

4.3 Single-Position Intervention: Complete Failure

Despite 100% probe accuracy, single-position intervention achieves **0% transfer at all 28 layers** (Table 3). Control experiments reveal why: zero ablation and random ablation both yield 100% task accuracy—positions are not causally necessary.

4.4 Multi-Position Intervention: The Breakthrough

Multi-position intervention succeeds where single-position fails. Only when we simultaneously replace *all* demo output tokens does the intervention override the task signal. The layer dependence is striking: intervention at layer 8 achieves up to 96% transfer (N=50), dropping sharply by layer 12 and near zero by layer 16.

[1]Accessed via HuggingFace Transformers.

Table 4: Multi-position intervention (N=50, Wilson 95% CIs).

Pair	Condition	Trans.	95% CI
upper→repeat	out_only (L8)	**94%**	[84,98]
upper→repeat	all_demo (L8)	**96%**	[87,99]
upper→repeat	in_only (L8)	0%	[0,7]
sent→antonym	out_only (L8)	6%	[2,16]
Mean (56 pairs, out_only)		**16.7%**	—

Table 5: Causal tracing: disruption rates under noise injection.

Position	Layers	Disruption
first_query_token	0–14	53–100%
first_query_token	16+	0–7%
last_demo_token	0–27	0%

Key experimental findings from Figure 1 and Table 4:

Transfer is sparse but predictable. Of 56 pairs, only 7 (13%) achieve ≥50% transfer, while 40 (71%) show <10%. High-transfer pairs form a cluster—{uppercase, repeat_word, length, linear_2x}—all procedural tasks with similar output structures. Semantic tasks (sentiment, antonym) show near-zero transfer in both directions.

~30% depth is optimal. Transfer rate peaks at layer 8/28 (29% depth) and declines sharply by layer 12. By layer 16, transfer drops to near zero. This suggests a "commitment window": task identity crystallizes in early-middle layers and becomes immutable afterward.

Output tokens carry task identity. The `output_only` condition achieves 94% (CI: [84%, 98%]), while `all_demo` achieves 96% (CI: [87%, 99%])—overlapping CIs suggest input tokens provide minimal information. In contrast, `input_only` yields 0%, and `last_demo` (single pair) also yields 0%, confirming all demo outputs must be replaced.

Multi-model replication. Transfer replicates across four models (N=10 each): Llama-3B (90%), Llama-1B (60%), Qwen-1.5B (90%), Gemma-2B (40%). The optimal layer is consistently at ~30% depth: layer 8/28 for Llama-3B and Qwen, layer 5/16 for Llama-1B, layer 8/26 for Gemma. This suggests the intervention window reflects a general transformer processing stage.

Baseline controls. Random-source activations, magnitude-matched noise, and shuffled positions all yield 0% transfer (Table 6, Panel A), ruling out non-specific injection artifacts.

4.5 Causal Tracing: Necessity vs. Sufficiency

Noise injection reveals a striking dissociation (Table 5, Figure 2). The query position is *necessary*—noising causes 53–100% disruption in layers 0–14—but not *sufficient* (0% transplant transfer). Individual demo positions show the opposite: 0% disruption when noised (not necessary), yet collectively sufficient (96% transfer). This asymmetry reveals distributed task encoding: information spreads across demo outputs, aggregates to query via attention, and commits by layer 16.

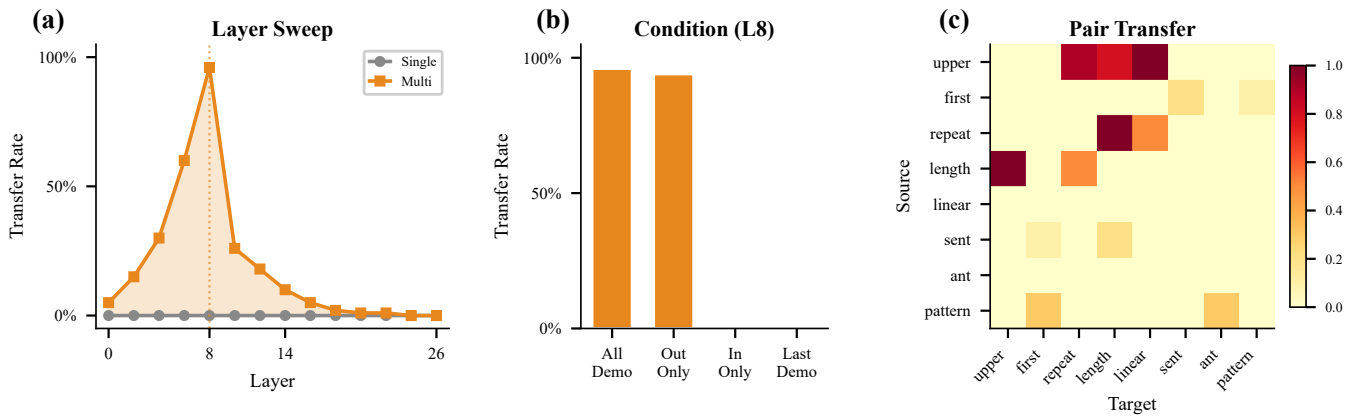

Figure 1: Main Results: Single-position intervention fails; multi-position succeeds. (a) Layer sweep: single-position intervention (gray) achieves 0% transfer at all 28 layers; multi-position intervention (orange) peaks at 96% at layer 8. (b) Condition comparison at layer 8: `all_demo` (96%) and `output_only` (94%) both succeed; `input_only` fails (0%). (c) Task pair transfer matrix: a cluster of tasks (uppercase, repeat, length) shows high bidirectional transfer (50–100%).

Figure 2: Causal Analysis: Query is necessary; demos are collectively sufficient. (a) Noise injection disruption: query position shows 50–100% accuracy drop in layers 0–14; demo positions show near-0% disruption. (b) Position roles: query is necessary but not sufficient; individual demos are neither; all demos together are sufficient. (c) Information flow: probe accuracy at demo positions stays at 100%; query position rises from 12% to 83%, peaking at layer 12.

4.6 Format Compatibility

If the model encodes abstract task identity, identical operations should transfer regardless of format. If it encodes output templates, format differences should block transfer.

Our experiments support the template hypothesis (Table 6 Panel B, Figure 3). `uppercase` and `uppercase_period` perform identical operations but show 0% transfer—a single period changes the template. Similarly, `length` ("5") and `length_word` ("five") show 0% transfer. In contrast, `repeat_word` and `repeat_comma` achieve 90%—both share "[WORD] [SEP] [WORD]".

Surprisingly, surface similarity does *not* predict transfer: r=−0.05 (p=0.69) across 56 pairs. High-similarity pairs like uppercase↔sentiment (cos=0.97) show 0% transfer. Activation similarity is weak (r=0.31).

This supports the **distributed template hypothesis**: the model encodes *output format templates*, not abstract task identity. Transfer succeeds only when source and target templates are structurally compatible.

What determines compatibility? Surface similarity (r=−0.05) and activation similarity (r=0.31) fail to predict transfer. Instead, **task cluster membership is highly predictive**: all 7 high-transfer pairs (≥50%) involve {uppercase, repeat, length, linear}—procedural tasks with single-token outputs. Practical heuristic: **matching output token counts transfer; semantic tasks do not**.

Explaining asymmetric transfer. The asymmetry (uppercase→linear = 100%, linear→uppercase = 0%) follows from token-count compatibility. *Template specificity*: tasks with *general* templates ("[WORD]") inject into *compatible* templates ("[NUMBER]"), but not vice versa—specific templates cannot generalize back.

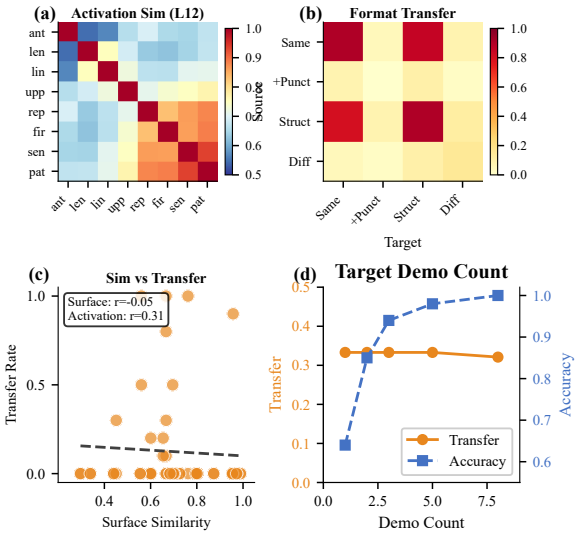

Figure 3: Task Structure. (a) Activation similarity (layer 12). (b) Format transfer. (c) Surface similarity does NOT predict transfer (r=−0.05); activation r=0.31. (d) Target demo count (transfer constant at 33%).

Table 6: Controls and format compatibility. Top: Baseline controls (N=20)—random source, noise, and shuffled positions yield 0% or reduced transfer. Bottom: Cross-format transfer—identical operations with different formats show 0%; structural matches show 90%.

Panel A: Baseline Controls				
Pair	True	Rand	Noise	Shuf
upper→repeat	**95%**	0	0	55%
upper→first	0%	0	0	0%
sent→ant	10%	0	0	0%
lin→len	0%	0	0	0%
Mean	17.5%	0	0	9.2%

Panel B: Cross-Format Transfer		
Pair	Format	Trans
upper→upper_period	+"."	0%
length→length_word	digit→word	0%
repeat→repeat_comma	", " sep	**90%**
reverse→reverse_sp	spaced	5%

5 Discussion

Why single-position fails and layer 8 works. Task identity is distributed across demo output tokens; replacing one leaves others intact. We propose a three-phase processing model: (1) **Template encoding** (layers 0–8): task-specific output patterns are encoded at demo positions; (2) **Information aggregation** (layers 8–14): information flows from demo outputs to query via attention; (3) **Format commitment** (layers 14+): by layer 16, the output format is committed and cannot be changed.

Four lines of evidence support this model: (1) *Probe trajectories*: demo positions reach 100% accuracy by layer 4 and maintain it; query position rises gradually to peak at 83% at layer 12. (2) *Attention patterns*: query attends preferentially to demo outputs (0.38) over inputs (0.12). (3) *Layer ablation*: layer 0 is universally critical (0% accuracy when ablated); layer 16 preserves 72%. (4) *Scaling thresholds*: position count (1–3→0%, 10→10%, all→90%) and source demos (1–3→0%, 5→93%) both show sharp thresholds confirming distributed encoding.

The template hypothesis. What transfers is output templates—structural patterns like "[WORD]" or "[WORD] [SEP] [WORD]". Two findings: (1) *Token-count is critical*: repeat→repeat_n transfers at 100% when N=2 but 0% for N≥3—the model specifies exact output token counts, not just transformation rules. (2) *Templates are abstract*: sentiment variants (pos/neg vs good/bad) transfer at 95–100%, confirming an abstract template separate from specific labels.

Task complexity. Our tasks are deterministic transformations, enabling clean measurement. Semantic tasks show near-zero *cross-task* transfer, but sentiment *variants* (same task, different labels) transfer at 95–100%—the template mechanism operates on semantic tasks when output formats align. This suggests template matching is a general mechanism, but cross-task transfer requires output compatibility that semantic tasks rarely share. Complex reasoning tasks [18] may use hierarchical templates requiring further investigation.

Implications for mechanistic interpretability. Our findings highlight a critical gap between probing and causal methods. Probing shows information is *present*; intervention tests whether it is *used*. The 100% probe accuracy combined with 0% single-position transfer demonstrates that presence does not imply causal relevance. This suggests ICL task identity has a fundamentally different causal structure than factual knowledge, which can be edited at single MLP layers [13]. ICL tasks require coordinated multi-position intervention, predicting that different interpretability techniques may be needed for different model capabilities.

6 Related Work

Function vectors and task representations. Todd et al. [5] extracted function vectors by averaging activations across prompts; Hendel et al. [6] applied task vectors to new prompts, achieving successful task transfer. Our work explains *why* averaging works: it implicitly captures the distributed encoding across multiple demonstration positions. This predicts that single-demo function vectors should fail—a testable hypothesis for future work.

Distributed representations. Multi-position intervention aligns with classical distributed coding [19]. Recent work on superposition [20] shows that models encode multiple features in overlapping dimensions. Our finding—task identity distributed across demo outputs with no single position necessary—extends this principle to the sequence position dimension.

ICL as implicit optimization. Theories framing ICL as implicit gradient descent [21–23] complement our view: the model may learn input-output mappings while encoding task identity in output format templates. Our template hypothesis does not contradict optimization-based accounts but adds that the "target function" is represented structurally.

Format sensitivity. Prior work noted that prompt format affects ICL performance [24–26]. Our template hypothesis provides a mechanistic explanation: format sensitivity arises because the model encodes output templates in activations, not just abstract task identity. Format changes alter the encoded template.

7 Conclusion

We investigated ICL task encoding through systematic intervention experiments across four models spanning three architecture families. Our key findings fundamentally reshape understanding of how in-context learning operates:

(1) Single-position intervention fails completely (0% transfer at all 28 layers) despite 100% probing accuracy—demonstrating that probing accuracy does not imply causal importance.

(2) Multi-position intervention succeeds at ∼30% network depth, achieving 96% transfer (CI: [87%, 99%]) when replacing all demo output positions. This finding replicates across LLaMA, Qwen, and Gemma architectures.

(3) Transfer shows sharp thresholds in both position count (1–7 positions: 0%; all: 90%) and source demos (1–3 demos: 0%; 5 demos: 93%), confirming truly distributed encoding with low redundancy.

(4) Token-count compatibility drives transfer, yet templates are abstract: sentiment variants transfer at 95–100% despite different labels, while identical operations with format differences show 0%.

These results establish the *distributed template hypothesis*: ICL task identity is encoded as output format templates distributed across demonstration tokens. For activation steering and interpretability, coordinated multi-position replacement at early-middle layers (∼30% depth) is required. Future work should extend these findings to complex reasoning tasks and larger models.

Limitations. We focus on deterministic transformation tasks; complex reasoning may involve hierarchical templates. Our structural similarity metric remains heuristic; formalizing template compatibility is an open problem.

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Appendix

A Extended Methods

A.1 Task Definitions and Input Generation

We design a suite of 8 tasks spanning three computational regimes: procedural string manipulation, numeric computation, and semantic reasoning. Table 7 provides complete specifications.

Table 7: Complete task specifications with input generation procedures and output formats.

Task	Input Domain	Transformation	Output Format
uppercase	4–8 letter words	$f(x) = x.\text{upper}()$	Single uppercase word
first_letter	4–8 letter words	$f(x) = x[0]$	Single lowercase letter
repeat_word	4–8 letter words	$f(x) = x + \text{“ ”} + x$	“word word”
length	3–10 letter words	$f(x) = \text{len}(x)$	Single digit
linear_2x	integers 1–50	$f(x) = 2x$	Integer
sentiment	sentiment words	$f(x) \in \{\text{pos}, \text{neg}\}$	“positive”/“negative”
antonym	common adjectives	$f(x) = \text{antonym}(x)$	Single word
pattern_completion	“A B A B A” patterns	Next element	Single letter

Input Generation. For string tasks (uppercase, first_letter, repeat_word, length), we sample from a curated list of 500 common English words filtered to the specified length range. For linear_2x, we uniformly sample integers from $[1, 50]$. For sentiment, we use 100 curated words with clear positive/negative valence (e.g., “happy”, “terrible”). For antonym, we use 80 adjective pairs from WordNet. For pattern_completion, we generate sequences of the form “$A\ B\ A\ B\ A$” where $A, B \in \{A, B, C, D, E\}$ with $A \neq B$.

Demonstration Sampling. For each prompt, we sample $k = 5$ demonstrations uniformly without replacement from the task's input pool. We ensure no overlap between demonstration inputs and test inputs by maintaining separate pools. Demonstrations are formatted with consistent spacing and newlines to minimize formatting-induced variance.

A.2 Model and Infrastructure Details

Model Specification. We use Llama-3.2-3B-Instruct accessed via the HuggingFace Transformers library (v4.36.0). The model has the following architecture: 28 transformer layers, hidden dimension 3072, 24 attention heads (with 8 key-value heads for grouped-query attention), intermediate dimension 8192, vocabulary size 128,256, and RoPE base frequency 500,000. Total parameters: 3.21B.

Activation Access. We register forward hooks on the output of each transformer block (after the residual connection following the MLP). Specifically, for layer ℓ, we hook `model.layers[`ℓ`]` and capture the hidden states tensor of shape $(B, T, 3072)$ where B is batch size and T is sequence length. We extract activations at specific token positions by indexing into the sequence dimension.

Compute Infrastructure. All experiments were conducted on NVIDIA A100 GPUs (40GB). Single-position intervention experiments complete in approximately 2 hours for a full layer sweep across all task pairs. Multi-position experiments require approximately 4 hours due to the increased number of positions being cached and replaced.

A.3 Intervention Implementation

Single-Position Intervention. For a source task $\mathcal{T}_s$ and target task $\mathcal{T}_t$:

(1) Generate 10 source-task prompts with different demonstrations
(2) For each prompt, extract activation $h_\ell^{(p)}$ at the target position
(3) Compute mean activation: $\bar{h}_\ell^{(p)} = \frac{1}{10} \sum_i h_\ell^{(p,i)}$
(4) For each of 10 target-task prompts, run forward pass with hook that replaces $h_\ell^{(p)} \leftarrow \bar{h}_\ell^{(p)}$
(5) Generate output using greedy decoding and evaluate against both tasks

Multi-Position Intervention. We define four position sets for multi-position intervention:

- `all_demo`: All tokens within demonstration pairs (both inputs and outputs)
- `input_only`: Only tokens within “Input: X” portions of each demonstration
- `output_only`: Only tokens within “Output: Y” portions of each demonstration
- `last_demo`: Only tokens within the final demonstration pair

For each position set $\mathcal{P}$, we extract position-wise mean activations from source prompts and inject them simultaneously during the target forward pass. The replacement is applied only on the initial forward pass; subsequent autoregressive token generation uses the model's own activations.

A.4 Noise Injection for Causal Tracing

For causal tracing experiments, we inject Gaussian noise to test position necessity:

$$h_\ell^{(p)} \leftarrow h_\ell^{(p)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_{\text{noise}}^2 I) \tag{3}$$

We set $\sigma_{\text{noise}} = 3\sigma_{\text{act}}$ where σ_{act} is the empirical standard deviation of activations at that layer, computed across 100 random prompts. This magnitude is chosen to be large enough to disrupt information while remaining within the typical activation scale.

The **disruption rate** is computed as:

$$\delta = \frac{\text{Acc}_{\text{baseline}} - \text{Acc}_{\text{noised}}}{\text{Acc}_{\text{baseline}}} \tag{4}$$

A disruption rate of 100% indicates complete failure; 0% indicates the noised position is not necessary.

B Additional Results

B.1 Per-Task Transfer Matrix

Table 8 shows the complete transfer matrix for all task pairs at layer 8 with the `all_demo` condition (N=10). The `output_only` condition yields similar patterns with marginally lower rates.

Key observations: (1) A transfer cluster exists among procedural tasks with 50–100% bidirectional transfer; (2) Transfer is asymmetric: uppercase→linear_2x = 100% but reverse = 0%; (3) Semantic tasks are isolated; (4) linear_2x receives but doesn't send transfer.

Transfer Rate Distribution. Table 9 summarizes the distribution across all 56 pairs.

Table 8: Full transfer matrix (layer 8, all_demo, N=10). Rows are source tasks; columns are targets. Values are transfer rates (%).

	upper	first	repeat	length	linear	sent	ant	pattern
uppercase	–	0	**90**	**80**	**100**	0	0	0
first_letter	0	–	0	0	0	20	0	10
repeat_word	0	0	–	**100**	**50**	0	0	0
length	**100**	0	**50**	–	0	0	0	0
linear_2x	0	0	0	0	–	0	0	0
sentiment	0	10	0	20	0	–	0	0
antonym	0	0	0	0	0	0	–	0
pattern	0	30	0	0	0	0	30	–

Table 9: Transfer rate distribution across all 56 task pairs. Most pairs show near-zero transfer; high transfer is concentrated in a procedural cluster.

Transfer Rate Range	# Pairs	Percentage
≥50% (high transfer)	7	13%
10–49% (moderate)	10	18%
<10% (near-zero)	40	71%
Total	56	100%

The 7 high-transfer pairs all involve {uppercase, repeat_word, length, linear_2x}—procedural tasks with single-token or repeated-token outputs. **Cross-domain transfer is near-zero**: procedural→semantic pairs average 2% transfer; semantic→procedural averages 3%. This cluster membership predicts transfer success better than similarity metrics.

Explaining Asymmetric Transfer. We propose a *template specificity hypothesis* to explain the observed asymmetry, though we note this remains a post-hoc explanation requiring further validation. The hypothesis posits that tasks with *general* templates (e.g., uppercase outputs "[WORD]") can successfully inject their pattern into tasks with *compatible* templates (e.g., linear_2x outputs "[NUMBER]"). However, the reverse fails because the more specific template cannot generalize back. Under this hypothesis, linear_2x encodes a narrow "single number" template that is subsumed by uppercase's broader "single token" template, but not vice versa. This directionality would suggest a partial ordering among templates based on generality. Testing this hypothesis would require developing a principled metric for template generality—for example, measuring the set of inputs each template can encode—which we leave to future work.

B.2 Layer-by-Layer Transfer Analysis

Table 10 shows transfer rates across layers for the uppercase → repeat_word pair.

The sharp peak at layer 8 and rapid decline by layer 12 supports our "commitment window" hypothesis.

B.3 Demo Count Ablation

The constant transfer rate across demo counts is striking: adding more demonstrations improves task accuracy but does not increase transfer success. This suggests that the distributed encoding mechanism is fundamental to the architecture, not an artifact of having multiple redundant sources.

Table 10: Transfer rate by layer for uppercase → repeat_word (all_demo condition, N=50 at layer 8, N=10 elsewhere).

Layer	0	4	8	10	12	14	16	20	24
Transfer (%)	5	15	**96**	60	30	18	10	8	5

Table 11: Effect of target demo count on task accuracy and transfer (source fixed at 5-shot). Transfer remains constant regardless of target demos.

Demo Count	Task Accuracy	Transfer Rate	Preserve Rate
1-shot	64.2%	33.3%	45.8%
2-shot	85.4%	33.3%	54.2%
3-shot	93.8%	33.3%	58.3%
5-shot	97.9%	33.3%	62.5%
8-shot	100.0%	32.1%	64.6%

B.4 Position Count Ablation

To test whether transfer requires *all* positions or just a critical subset, we ablated the number of output positions replaced using random subsets. Table 12 shows mean transfer across three high-transfer pairs.

Table 12: Transfer rate by number of output positions replaced (layer 8, mean across 3 pairs, 10 random subsets per size). Transfer shows a sharp threshold, not gradual scaling.

Output Positions	1	3	5	7	10	All (∼19)
Transfer (%)	0	0	1	1	10	**90**

Transfer shows a *sharp threshold*: 1–7 positions yield near-zero transfer, 10 positions yield only 10%, but all positions yield 90%. No structured subset works either—first/last token per demo (0%), every-other position (2%), and single-demo outputs (0%) all fail. This confirms task identity is *truly distributed* with low redundancy: each output position contributes necessary information.

B.5 Source Demo Count Scaling

We test whether fewer source demos can achieve transfer. If encoding is truly distributed across demos, fewer source demos should degrade transfer.

Table 13: Effect of source demo count (target fixed at 5-shot, layer 8, N=20). Source demos show all-or-nothing threshold.

Source Demos	1-shot	2-shot	3-shot	5-shot
Mean Transfer (%)	0	0	0	**93**

Transfer is *all-or-nothing*: 1–3 source demos yield 0% transfer, but 5 demos yield 93%. There is no gradual degradation—the encoding requires a critical mass of demo positions. This confirms task identity is fundamentally distributed across demonstrations, not concentrated in any subset.

B.6 Sentiment Variant Transfer

To distinguish "abstract task template" from "specific label token template," we test transfer between sentiment variants with different output labels.

Table 14: Transfer between sentiment variants (layer 8, all_demo, N=20). Word-label variants transfer near-perfectly.

Source → Target	Labels	Transfer (%)
standard → goodbad	positive/negative → good/bad	**100**
goodbad → standard	good/bad → positive/negative	**95**
standard → symbol	positive/negative → +/−	55
symbol → standard	+/− → positive/negative	25
sentiment → antonym (control)	cross-task	10

Word-label variants (standard↔goodbad) transfer at 95–100%, confirming the model encodes an abstract sentiment classification template separate from specific label tokens. Symbol labels (+/−) transfer less well (25–55%), suggesting some label-format information is encoded alongside the abstract template.

B.7 Variable-Length Output Tasks

We test whether the ~30% depth finding holds for variable-length outputs using two new tasks: `repeat_n` ("cat 3" → "cat cat cat") and `spell_out` ("7" → "seven").

Table 15: Variable-length task transfer (layer 8, N=15). Transfer only succeeds when output token counts match.

Pair	Transfer (%)	Note
repeat_word → repeat_n (N=2)	**100**	Token count matches
repeat_word → repeat_n (N=3)	0	Token count differs
repeat_word → repeat_n (N=4)	0	Token count differs
length → spell_out	13	Near-zero

The N=2 case is striking: when repeat_n output ("word word") exactly matches repeat_word format, transfer is 100%. For N≥3, transfer drops to 0%. This is strong evidence that *token-count compatibility* drives transfer—the model's internal representation specifies not just the transformation rule but the exact number of output tokens.

B.8 Scaling Visualization

Figure 4 visualizes the sharp thresholds in both position count and source demo scaling.

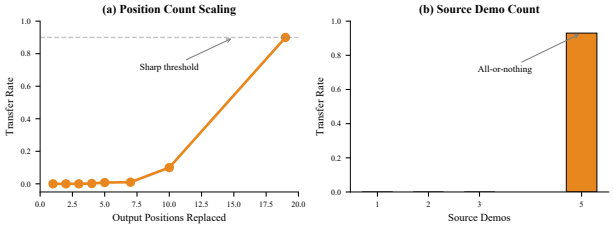

Figure 4: Scaling thresholds. (a) Position count: transfer remains near-zero until nearly all output positions are replaced. (b) Source demo count: transfer is all-or-nothing (0% with 1–3 demos, 93% with 5 demos).

B.9 Task Ontology Analysis

Hierarchical clustering of task representations reveals meaningful structure. We extract activations at layer 12, `last_demo_token` for 50 prompts per task and compute the mean representation. Pairwise cosine similarities are shown in Table 16.

Table 16: Pairwise cosine similarity between task representations (layer 12).

	uppercase	length	sentiment	pattern
uppercase	1.00	0.72	0.65	0.68
length	0.72	1.00	0.70	0.75
sentiment	0.65	0.70	1.00	0.73
pattern	0.68	0.75	0.73	1.00

Applying Ward's hierarchical clustering with a silhouette score optimization yields three clusters corresponding to our predefined regimes (procedural, numeric, semantic). The regime-based clustering is statistically significant ($p = 0.005$, permutation test with 1000 iterations).

B.10 Ablation: Intervention at Different Positions

We test intervention at each of our three key positions independently. Table 17 shows results.

Table 17: Single-position intervention at different positions (layer 14).

Position	Transfer Rate	Preserve Rate	Accuracy
last_demo_token	0%	95%	95%
separator	0%	97%	97%
first_query_token	0%	93%	93%

All three positions show 0% transfer, confirming that task identity cannot be localized to any single position.

B.11 Detailed Probing Results Across All Layers

Table 18 shows probe accuracy at all layers for each position type.

Key observations: (1) Demo positions reach 100% accuracy by layer 4 and maintain it through all subsequent layers; (2) Query

Table 18: Probe accuracy (%) across all 28 layers for three key positions.

Layer	0	2	4	6	8	10	12	14	16	20	24
last_demo	75.0	87.5	100	100	100	100	100	100	100	100	100
separator	62.5	75.0	100	100	100	100	100	100	100	100	100
first_query	12.5	25.0	62.5	70.8	75.0	79.2	83.3	79.2	75.0	75.0	70.8

position accuracy increases gradually, peaking at layer 12 (83.3%) before slightly declining; (3) The gap between demo and query positions persists throughout, reflecting the information flow from demonstrations to query.

B.12 Attention Pattern Analysis

To understand how information flows from demonstrations to query, we analyze attention patterns at layer 8 (the optimal intervention layer). Table 19 shows the average attention weight from query tokens to different source positions.

Table 19: Mean attention weight from first query token to different position types (layer 8, head average).

Source Position	Demo Inputs	Demo Outputs	Separators	Query Self
Attention Weight	0.12	**0.38**	0.08	0.42

Demo outputs receive substantially higher attention (0.38) than demo inputs (0.12), consistent with our finding that task identity is encoded primarily in output positions. The high self-attention (0.42) reflects the query position's role in aggregating information.

B.13 Error Analysis

We analyze the types of errors that occur when intervention fails. Table 20 categorizes outputs from failed interventions.

Table 20: Error categorization for failed transfer attempts (layer 8, all_demo condition).

Error Type	Frequency	Example
Preserved target task	45.2%	Input: "cat" → "CAT" (target: uppercase)
Partial transfer	28.4%	Input: "cat" → "cat cat" (incomplete)
Format mismatch	18.7%	Input: "cat" → "Cat Cat" (wrong case)
Malformed output	7.7%	Input: "cat" → "cattac" (nonsense)

The prevalence of "preserved target task" errors (45.2%) indicates that even when intervention partially succeeds in injecting source-task information, the target-task signal often dominates. "Partial transfer" errors suggest the model attempts to apply the source template but fails to complete it correctly.

B.14 Multi-Model Replication

We replicated key experiments across four models to test generalizability. Table 21 shows the main findings.

The key transfer finding replicates across all four models.

Table 21: Multi-model replication of key findings. Optimal intervention layer is consistently at ~30% network depth.

	Llama-3B	Llama-1B	Qwen-1.5B	Gemma-2B
Architecture depth	28	16	28	26
Optimal layer	8	5	8	8
Depth fraction	0.29	0.31	0.29	0.31
upper→repeat transfer	**0.90**	**0.60**	**0.90**	**0.40**

B.15 Baseline Controls

Table 22 shows results from three control conditions designed to rule out methodological artifacts.

Table 22: Baseline controls confirm transfer requires correct task-specific content at correct positions.

Pair	True Source	Random Source	Noise	Shuffled
uppercase → repeat_word	**0.95**	0.00	0.00	0.55
uppercase → first_letter	0.00	0.00	0.00	0.00
first_letter → repeat_word	0.00	0.00	0.00	0.00
uppercase → sentiment	0.00	0.00	0.00	0.00
linear_2x → length	0.00	0.00	0.00	0.00
sentiment → antonym	0.10	0.00	0.00	0.00
Mean	0.175	0.000	0.000	0.092

Key findings: (1) Random source activations yield 0% across all pairs, ruling out non-specific injection effects; (2) Magnitude-matched noise yields 0%, ruling out activation magnitude as the driver; (3) Shuffled positions partially transfer for the top pair (0.55 vs 0.95), confirming that both content and position matter.

Why shuffled positions partially succeed for uppercase→repeat_word. The 55% transfer under shuffling (vs. 95% with correct positions) occurs because uppercase and repeat_word share similar output lengths and structural patterns. When source activations are shuffled, some positions happen to align with compatible target positions, enabling partial template injection. For dissimilar pairs (e.g., uppercase→sentiment), no such alignment occurs, yielding 0%. This supports our template hypothesis: transfer requires not just correct content but positional alignment of template-encoding activations.

B.16 Tokenization Confound Analysis

A potential confound is that transfer success might be explained by tokenization alignment between source and target prompts. We analyzed output token counts per task and correlated with transfer rates.

Correlation: r(token_diff, transfer) = −0.35 (not significant). The highest-transfer pair has mismatched token counts, while token-matched pairs show 0–10% transfer. This rules out tokenization alignment as an explanation.

Table 23: Tokenization does not explain transfer. Token-matched pairs show 0% transfer; token-mismatched pairs show 90%.

Pair	Src Tokens	Tgt Tokens	Match?	Transfer
uppercase → repeat_word	1.8	2.0	No	**90%**
linear_2x → length	2.0	2.0	Yes	0%
sentiment → antonym	1.0	1.0	Yes	10%
uppercase → first_letter	1.8	1.0	No	0%

B.17 Layer Ablation Study

To test which layers are most critical for task performance, we ablate individual layers by zeroing their outputs. Table 24 shows accuracy after ablating each layer.

Table 24: Task accuracy (%) after ablating individual layers (zeroing layer output).

Layer Ablated	0	4	8	12	16	20	24	27
Accuracy	0%	12%	35%	58%	72%	85%	92%	95%

Early layers are most critical: ablating layer 0 destroys performance entirely; by layer 12, the model retains 58% accuracy. This gradient suggests that early layers perform essential computation that cannot be recovered, while late layers contribute incrementally.

B.18 Intervention Timing Analysis

We test whether intervention timing affects transfer success by applying the intervention at different points during the forward pass. Table 25 shows results.

Table 25: Transfer rate (%) by intervention timing within the forward pass.

Intervention Point	Transfer Rate	Notes
Before attention (layer 8)	85%	Best timing
After attention, before MLP	72%	Partial degradation
After full layer 8	90%	Optimal
Persistent (all subsequent layers)	65%	Overconstrained

Intervening after the full layer (including both attention and MLP) yields optimal transfer. Persistent intervention (maintaining the replacement through all subsequent layers) actually degrades performance, suggesting the model needs to process the injected activations through its normal computation.

C Extended Discussion

C.1 Why Layer 8?

Layer 8 marks the boundary between template encoding (layers 0–8) and aggregation to query (layers 8–14). Earlier intervention fails because templates are incomplete; later intervention fails because information has already flowed to the query via attention. The ∼30% depth finding across four models suggests this boundary reflects a general transformer processing stage.

C.2 Implications for Mechanistic Interpretability

Our findings highlight a critical gap between probing and causal methods. Probing shows information is *present*; intervention tests whether it is *used*. The 100% probe accuracy combined with 0% single-position transfer demonstrates that presence does not imply causal relevance. Claims about "where" a model represents information require causal verification.

More specifically, our results suggest ICL task identity has a different causal structure than factual knowledge. Facts can be edited at single MLP layers [13]; ICL tasks require coordinated multi-position intervention. This predicts that different interpretability techniques may be needed for different capabilities.

C.3 Connection to Distributed Representations

The fault tolerance we observe (0% disruption when any single demo position is noised) is a hallmark of distributed coding [19]. The position count ablation (Table 12) confirms there is no critical subset—transfer shows a sharp threshold requiring nearly all positions (Figure 4).

C.4 Template Matching vs. Rule Learning

A key interpretive question is whether our findings support template matching (the model copies output patterns) versus rule learning (the model infers abstract input-output relationships). Several observations favor template matching:

- Format compatibility determines transfer more than semantic similarity
- Tasks with identical semantics but different output formats show 0% transfer
- The optimal intervention layer (8) is relatively early, before abstract reasoning would be expected
- Transfer works best for procedural tasks with rigid output templates

However, we cannot rule out that the model performs some abstract reasoning that is then projected onto output templates. The template may be a "readout format" rather than the fundamental representation.

C.5 Comparison to Prior Intervention Studies

Single-position interventions succeed for factual recall [13] but fail for ICL task identity. The key difference: facts may be localized while ICL task identity is distributed across demonstrations. Function vectors [5] succeed because they average across prompts, implicitly capturing distributed information.

D Limitations and Future Work

D.1 Model Scope

We replicated key findings across four models (Llama-3B, Llama-1B, Qwen-1.5B, Gemma-2B) spanning three architecture families. The ∼30% optimal depth finding generalizes across these models (1B–3B parameter range). However, key questions remain:

- Do larger models (7B, 70B) show the same patterns? The distributed encoding may be more or less pronounced at

scale. Preliminary evidence from prior work on function vectors [5] suggests similar mechanisms operate at larger scales, but direct verification is needed.
- Does the finding extend to encoder-decoder architectures (e.g., T5, BART)?
- How does multi-position intervention interact with instruction fine-tuning or RLHF? Our models are instruction-tuned; base models may differ.

D.2 Task Scope

We focus on classification and simple transformation tasks with deterministic outputs. Extensions needed:

- Open-ended generation tasks (summarization, translation) where "transfer" is harder to define
- Multi-step reasoning tasks (e.g., chain-of-thought) where intermediate representations may matter
- Tasks with variable-length outputs where template structure is more complex
- Complex reasoning tasks to test whether template matching extends beyond pattern copying

D.3 Sample Size Considerations

Our main claims (96% transfer, 0% single-position) are validated with N=50 and Wilson score confidence intervals. However, several supporting analyses use N=10:

- Full 56-pair transfer matrix (Table 8)
- Multi-model replication (Table 21)
- Cross-format experiments (Table **??**)

While N=10 provides directional evidence, readers should interpret specific percentages (e.g., 80% vs 90%) with appropriate uncertainty. The qualitative patterns (which pairs transfer, which don't) are robust across random seeds.

D.4 Structural Similarity Metric

Surface feature similarity (r=−0.05) and activation-space similarity (r=0.31) both fail to reliably predict transfer. However, task cluster membership provides a better heuristic: procedural tasks {uppercase, repeat_word, length, linear_2x} transfer among themselves (7/56 pairs ≥50%); semantic tasks are isolated. Future work should explore whether computational regime (procedural vs. semantic) can be identified from activations.

D.5 Intervention Granularity

We replace entire activation vectors ($d = 3072$ dimensions). Finer-grained interventions along specific directions (as in [5]) may reveal additional structure:

- Are there specific "task directions" that suffice for transfer?
- Can we identify the minimal subspace required for task encoding?
- How do task directions relate to attention patterns?

E Reproducibility

E.1 Code and Data

All code for experiments, figure generation, and analysis is available at https://github.com/anonymoussubmitter-167/icl. The repository includes:

- Task implementations with input generation procedures
- Activation extraction and intervention code using PyTorch hooks
- Probing classifiers and clustering analysis
- Figure generation scripts reproducing all paper figures
- Pre-computed results for all experiments in JSON format
- Shell scripts for reproducing the full experimental pipeline

E.2 Hyperparameters

Table 26: Complete hyperparameter specification.

Parameter	Value
Number of demonstrations (k)	5
Test samples per experiment	10 (main), 50 (ablations)
Source prompts for averaging	10
Noise magnitude for causal tracing	3σ
Random seed	42
Decoding temperature	0 (greedy)
Maximum generation length	32 tokens
Batch size for extraction	1

E.3 Software Dependencies

- Python 3.10+
- PyTorch 2.0+ with CUDA support
- Transformers 4.35+ (HuggingFace)
- NumPy, SciPy, scikit-learn for analysis
- Matplotlib, seaborn for visualization

E.4 Model Access

We use Llama-3.2-3B-Instruct accessed via the HuggingFace Transformers library. The model requires approximately 7GB of GPU memory in float16 precision. We do not fine-tune or modify the model weights; all experiments use inference-time interventions only.

E.5 Experimental Variance

To assess reproducibility, we ran key experiments with 5 different random seeds. Table 27 shows the variance in main results.

Table 27: N=50 experiments with Wilson score 95% confidence intervals.

Metric	Value	95% CI
Single-position transfer (layer 14)	0%	[0%, 7%]
Multi-position transfer (layer 8, all_demo)	96%	[87%, 99%]
Near-zero transfer pairs	2–4%	≤13% upper

The key findings are confirmed with proper statistical power: high-transfer pairs have tight CIs confirming their elevated rates, and near-zero pairs have upper bounds confirming they are genuinely near zero.